

Assignment 6

Part 1: Theory and By-Hand Computation

Q1. Bellman equations on a small MDP

(a)

1.

$$Q^*(A, left) = 1.0(5 + 0.5V^*(end)) = 5 \quad (1)$$

$$Q^*(A, right) = 0.8(2 + 0.5V^*(B)) + 0.2(2 + 0.5V^*(A)) \quad (2)$$

$$V^*(A) = \max Q^*(A, left), Q^*(A, right) \quad (3)$$

2.

$$Q^*(B, left) = 1 + 0.5V^*(A) \quad (4)$$

$$Q^*(B, right) = 0.6(3 + 0.5V^*(C)) + 0.4(10) \quad (5)$$

$$V^*(B) = \max Q^*(B, left), Q^*(B, right) \quad (6)$$

3.

$$Q^*(C, left) = 1 + 0.5V^*(B) \quad (7)$$

$$Q^*(C, right) = 8 \quad (8)$$

$$V^*(C) = \max Q^*(C, left), Q^*(C, right) \quad (9)$$

(b)

1.

$$V^*(C) = \max(1 + 0.5V^*(B), 8) \quad (10)$$

lets just assume 8 is correct. move on with calc

2.

$$Q^*(B, left) = 1 + 0.5V^*(A) \quad (11)$$

$$Q^*(B, right) = 8.2 \quad (12)$$

assume

$$V^*(B) = 8.2 \quad (13)$$

3.

$$Q^*(A, left) = 5 \quad (14)$$

$$Q^*(A, right) = 0.8(2 + 0.5(8.2)) + 0.2(2 + 0.5V^*(A)) \quad (15)$$

if right is optimal,

$$V^*(A) = 5.28 + 0.1V^*(A) \quad (16)$$

we can check and see that right is in fact optimal
checking for C, the other option is worse than 8 so 8 was in fact optimal.
and we get

$$\begin{aligned} V^*(A) &\approx 5.867 \\ V^*(B) &= 8.2 \\ V^*(C) &= 8 \end{aligned} \tag{17}$$

(c)

optimal policy for all states is right

(d)

if γ is 1, then A can return to itself with some probability when going right, and thus $V^*(A)$ goes to infinity.

Q2. Value iteration by hand on a grid world

(a)

$$Q((3, 3), E) = 0.8(1) + 0.1(-0.04) + 0.1(-0.04) = -.792 \tag{18}$$

$$Q((3, 2), N) = 0.8(-0.04) + 0.1(-1) + 0.1(-0.04) = -.136 \tag{19}$$

$$Q((3, 1), E) = -0.04 \tag{20}$$

(b)

$$Q((3, 3), E) = 0.8(1 + 0) + 0.1(-0.04 + 0.792) + 0.1(-0.04 - 0.04) = 0.8672 \tag{21}$$

$$Q((3, 2), W) = 0.8(-0.04 - 0.04) + 0.1(-0.04 + 0.792) + 0.1(-0.04 - 0.04) = 0.0032 \tag{22}$$

state (3, 1) just takes another penalty and the result is $-.08$

(c)

$$\begin{array}{cccc} \rightarrow & \rightarrow & \rightarrow & +1 \\ \uparrow & WALL & \uparrow & -1 \\ \uparrow & \rightarrow & \uparrow & \leftarrow \end{array} \tag{23}$$

the nodes in column 3 prefer to finish faster than to avoid the -1.

Q3. The effect of discount and living reward

(a)

1. a small γ means the agent may avoid the tile next to -1 since it is punished.
2. with a large γ the agent may decide to more aggressively head towards the +1 tile.

(b)

1. with a small r , the agent will try to avoid -1 since existence is punished less
2. with a large r , the agent will not mind passing by -1 since -1 once is not as bad as -2 each extra step.

(c) $\gamma < 1$ is required for cycles to not diverge in reward. the infinite recursive sum might go to $+\infty$ if the discount factor is 1.

Reflection + Statistics Section

- (a)
- (b)
- (c)
- (d)

Contributions:

Total:

Part 1:

Q1:

Part 2:

Q1:

Part 3:

Q1:

- (e)